

Rohit Shah

Senior Data scientist

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Founder: [Mastering LLM](#) | [Linked-in](#)

PROFILE SUMMARY

- Senior data scientist at Fractal Analytics with 11+ years of experience leading insurance vertical and AI COE for Gen AI.
- Worked with Marshall Goldsmith to build his [digital twin](#).
- Built Chat with your data (agentic RAG) for the largest insurance company (fortune 100) currently used by ~300 leadership team & planned to rollout across organization for 24k employee.
- Built a knowledge bot with open-source models for the largest insurance company on website data.
- Lead a project that helps 220 Customer support teams which will reduce 15% avg. Handling Time for Canada-based insurance client.
- Founder of page [Mastering LLM](#) with 20k+ active learners across the platform.
- Lead projects that provided a total saving of \$1 million for leading claim negotiation customers.
- Pivotal role in development of clinical NLP product which extracts clinical entity from medical notes.
- Proficient in design & development
 - **Ideation**- Adept at translating business problems into technology problems, project planning, rapid prototyping
 - **Software development** - Architecture design & hands-on experience in machine learning with different set of problems like Classification, Regression, Natural language processing (NLP), Time Series, Image
 - **Machine learning operationalization** - Drift detection, experiment tracking, CI/CD
- Hands-on experience in cloud services like GCP, AWS & Databricks for machine learning and big data analytics.

TECHNICAL SKILLS

Environments	Windows, Linux, Cloud (GCP, AWS, Azure, Data bricks), Oracle DB, Git, GitHub
Languages	Python, SQL, PL/SQL, Core Java, Unix, PySpark
ML & DL Models	Large Language models, Deep learning (DNN, CNN, RNN), BERT, Random Forest, XG boost, KNN, SVC, Statistical Modelling, Probability techniques, Statistics
Libraries	Hugging face, TensorFlow, Keras, pytorch, Scikit learn, Pandas, NumPy, Tensor Board, Matplotlib, Seaborn, google facets, LIME, SHAP, what-if tool, NLTK, Genism, Spacy, scispacy, hugging face, ML Flow, TFX, Hyperopts, fb prophet, wandb

Professional Experience

- **Chat with your own data (CWYD)**
 - Developed Agentic RAG based system to build CWYD capable of answering question (RAG), document summarization & comparison, chat with structured data & code generation with ~80% accuracy with support of multiple file formats like PDF, Excel, word, PPT, text.
 - Currently used by 300 leadership team and planned to scale it to 24k employee across organization by end of this year.
- **Project Leader and AI Researcher**
 - Developed chatbot with an open-source ecosystem on website data with 80% faithfulness, 90% answer relevancy & 88% accuracy on negative examples.

- Fine-tuned LLAMA 2 7B with QLORA with synthetic data which gave comparable results to LLAMA 2 70B chat model.
- Developed decision framework for Large Language Model (LLM) systems and data strategies.
- Developed strategy to create high-quality synthetic data, fine-tuning and evaluating LLM system.
- Cost-benefit analysis for SaaS vs Open source, RAG vs fine-tuning & SLM vs LLM.
- Delivered 50+ hours of training sessions, including a deep dive for leadership team members.
- **CCC Support bot**
 - Client support team maintains QNs for specific sponsors - a document with multiple pages that has information about the contract details.
 - Developed dynamic prompting-based RAG system using AWS Tech stack with 82% accuracy to make retrieval of information easier which will reduce the 15% avg. handling Time.
- **Leading AI COE**
 - Led AI COE with a focus on developing reusable framework & accelerator to reduce turnaround time for client implementation.
 - Managed team of 25+ member across all projects and internal initiatives.
 - Engaged in over 20+ client engagements and partnership calls with major clients, including big pharmaceutical companies, financial institutions, and tech companies.
- **Digital Twin**
 - Worked with [Marshall Goldsmith](#) to build his digital twin with 96% accuracy.
 - Fine-tuned curie model as open Q&A to get nuances in the language and style of Marshall Goldsmith.
- **Price Optimization:** -
 - Reduce price discrepancy across 26 European countries such that turnover remains the same & risk is reduced due to a lack of market coordination.
 - Used evolutionary-based constrain optimization model to reduce risk by ~5% for 4 different objective functions.
- **Network Provider matching:**
 - 6 million claims per year requires manual matching which cost around \$1 million.
 - Developed random forest model which automates 60% of no match claim with ~1.7% of error rate which is less than human error rate of ~4% reported by business.
 - Same model is used as recommendation system for matched claims with 90% accuracy in top 5 recommendation.
- **Clinical NLP accelerator:**
 - The product is based on machine learning and NLP helped to abstract data locked in structure/unstructured notes (Medical records/Charts).
 - Model processes clinical notes in less than 1 sec to extract multiple entities with attributes with F1 score of 82%.
- **Medical code matching:**
 - Improved code set mapping efficiency (400%) from 15 concepts/hr to 60 concepts/hr through recommendation system for client specialized in medical vocabulary.
 - Two ML model (KNN) was developed for problem (95% accuracy in top 5 recommendation) and procedures (92% accuracy in top 5 recommendation), Features was created using Clinical BERT.
- **Claims Denial Management:**
 - XGboost model was developed with 95% precision to predict claims denied due to medical necessity prior to submission to payers.
 - Dataset was highly imbalance where only 1% of data belongs to positive class.
- **Machine learning model monitoring Framework:**
 - Managed 4-member cross functional team to create model monitoring framework for HIMSS to monitor evaluation metrics, business outcome, drift detection, model explanation & Infrastructure monitoring.

Machine learning competition & open-source contribution

- Predicting probability of success of new grocery store (1st price)
- CitiusTech Hackathon- Decentralized AI (1st price)
- Capgemini Tech Challenge (3rd position)
- Predicting promotional probabilities (Top 3%)
- Raised an issue with sci-kit learn for SVC model with degree three which is accepted as bug

- Reported issue for what-if tool (developed by google brain team) which was accepted as bug
- For more [projects & hackathon](#)

Profile

- **Portfolio:** <https://rshah1990.github.io/>
- **GitHub:** <https://github.com/rshah1990>
- **Kaggle:** <https://www.kaggle.com/rshah1990>
- **Analytics Vidhya:** <https://datahack.analyticsvidhya.com/user/profile/shah27>

Course undertaken

- Large Language Models: Application through Production (LLM101x Databricks)
- Large Language Models: Foundation Models from the Ground Up (LLM102x Databricks)
- GCP machine learning engineer, Year 2021
- GCP Data engineer (Coursera & Linux Academy), Year 2020
- Scalable machine learning on big data using Apache Spark (Coursera), Year 2020
- Data camp certificated Data scientist, Year 2016
- Oracle SQL & PL/SQL developer, Year 2014
- DEV288x: Natural language processing by Microsoft (Advance NLP)

Work History

Employer	Title	Dates of employment
Fractal Analytics	Senior Data scientist	Oct 2021 - Present
CitiusTech	Assistant manager of Data science	May 2019 – Oct 2021
Deloitte USI, Mumbai, India	Consultant (Oracle ERP & ML developer)	Feb 2017 - May 2019
Tata Consultancy services	Oracle Apps Developer	Mar 2013 – Feb 2017

EDUCATION

Bachelor of Engineering from Mumbai University in the year 2012, India.